ERTH555 Problem Set 1

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Problem 1

$$\rho \frac{\partial v_z}{\partial z} = -\frac{\partial P}{\partial z} + \eta \frac{\partial^2 v_z}{\partial x^2} + \rho g \tag{1}$$

a. Show that this equation is dimensionally homogeneous

To show that equation (1) is dimensionally homogeneous, we start by identifying the terms and their dimensions:

- ρ - material density $\{ = \}$ $\frac{M}{L^3}$
- η - viscosity $\{ = \}$ $\frac{M}{LT}$
- g - acceleration of gravity $\{ = \}$ $\frac{L}{T^2}$
- x - horizontal coordinate $\{ = \}$ L
- z - vertical coordinate $\{ = \}$ L
- t - time$\{ = \}$ T
- $v_z(x,t)$ - component of velocity in z direction $\{ = \}$ $\frac{L}{T}$
- $P(z,t)$ - pressure $\{ = \}$ $\frac{M}{LT^2}$
- $\frac{\partial}{\partial x}$ - partial derivative in x direction $\{ = \}$ $\frac{1}{L}$
- $\frac{\partial}{\partial z}$ - partial derivative in z direction $\{ = \}$ $\frac{1}{L}$
- $\frac{\partial^2}{\partial x^2}$ - 2nd partial derivative in x direction $\{ = \}$ $\frac{1}{L^2}$
- $\frac{\partial}{\partial t}$ - partial derivative in t direction $\{ = \}$ $\frac{1}{L^2}$

Now we can show that each term in (1) has the same dimensionality:

$$\rho\frac{\partial v_z}{\partial z}\{=\}\frac{M}{L^3}\times\frac{1}{T}\times\frac{L}{T}=\frac{M}{L^2T^2} \tag{2}$$
$$-\frac{\partial P}{\partial z}\{=\}1\times\frac{1}{L}\times\frac{M}{LT^2}=\frac{M}{L^2T^2} \tag{3}$$
$$\eta\frac{\partial^2 v_z}{\partial x^2}\{=\}\frac{M}{LT}\times\frac{1}{L^2}\times\frac{L}{T}=\frac{M}{L^2T^2} \tag{4}$$
$$\rho g\{=\}\frac{M}{L^3}\times\frac{L}{T^2}=\frac{M}{L^2T^2} \tag{5}$$

Since terms (2) - (5) all have dimension $\frac{M}{T^2L^2}$, the equation is dimensionally homogeneous.

b. Identify the independent variables, the dependent variables, and the constants.

The independent variables in (1) are: x, z, and t.

The dependent variables are: v_z and P since they can be written as functions of the independent variables.

The constants are ρ, η, and g since they are constant throughout the domain. The length w, and max v_z are also constant.

c. Define a set of dimensionless independent and dependent variables and dimensionless differential operators, and substitute them into the equation.

The set of dimensionless independent and dependent variables and differential operators are given by:

Spatial Variables:

$$\tilde{x}=\frac{x}{w}$$
$$\tilde{z}=\frac{z}{w}$$
$$\frac{\partial}{\partial\tilde{x}}=\frac{\partial x}{\partial\tilde{x}}\frac{\partial}{\partial x}=\frac{w}{1}\frac{\partial}{\partial x}$$
$$\frac{\partial^2}{\partial\tilde{x}^2}=(\frac{\partial x}{\partial\tilde{x}})^2\frac{\partial^2}{\partial x^2}=\frac{w^2}{1}\frac{\partial^2}{\partial x^2}$$
$$\frac{\partial}{\partial\tilde{z}}=\frac{\partial z}{\partial\tilde{z}}\frac{\partial}{\partial z}=\frac{w}{1}\frac{\partial}{\partial z}$$

where w is the width of the dike.

Velocity

$$\tilde{v_z} = \frac{v_z}{v_{z,max}}$$

where $v_{z,max}$ is the maximum vertical velocity occurring at x=0.

Time

$$\tilde{t} = \frac{t}{t^*}, t^* = \frac{w}{v_{z,max}}$$
$$\frac{\partial}{\partial \tilde{t}} = \frac{\partial t}{\partial \tilde{t}} \frac{\partial}{\partial t} = \frac{t^*}{1} \frac{\partial}{\partial t}$$

where $t^* = \frac{w}{v_{z,max}}$ is the time to move a the critical length w in the z direction.

Pressure

$$\tilde{P} = \frac{P}{\rho v_{z,max}^2}$$

where $\rho v_{z,max}^2$ is the dynamic pressure.

We can substitute these into (1) and get:

$$\rho \frac{\partial v_z}{\partial z} = -\frac{\partial P}{\partial z} + \eta \frac{\partial^2 v_z}{\partial x^2} + \rho g \tag{6}$$
$$= \rho(\frac{v_{z,max}}{w}) v_{z,max} \frac{\partial \tilde{v_z}}{\partial \tilde{z}} = -\rho v_{z,max}^2 (\frac{1}{w}) \frac{\partial \tilde{P}}{\partial \tilde{z}} + \eta (\frac{1}{w^2}) v_{z,max} \frac{\partial^2 \tilde{v_z}}{\partial \tilde{x}^2} + \rho g \tag{7}$$
$$= (\frac{\rho v_{z,max}^2}{w}) \frac{\partial \tilde{v_z}}{\partial \tilde{z}} = -(\frac{\rho v_{z,max}^2}{w}) \frac{\partial \tilde{P}}{\partial \tilde{z}} + (\frac{\eta v_{z,max}}{w^2}) \frac{\partial^2 \tilde{v_z}}{\partial \tilde{x}^2} + \rho g \tag{8}$$

d. Rearrange the groups of constants in the normalized equation so that all the terms are dimensionless and, in doing so, identify any dimensionless groups

Since our variables and operators are dimensionless, we can move the coefficients on the left hand side over, making the entire expression in (8) dimensionless.

$$(\frac{\rho v_{z,max}^2}{w}) \frac{\partial \tilde{v_z}}{\partial \tilde{z}} = -(\frac{\rho v_{z,max}^2}{w}) \frac{\partial \tilde{P}}{\partial \tilde{z}} + (\frac{\eta v_{z,max}}{w^2}) \frac{\partial^2 \tilde{v_z}}{\partial \tilde{x}^2} + \rho g \tag{9}$$
$$= \frac{\partial \tilde{v_z}}{\partial \tilde{z}} = -\frac{\partial \tilde{P}}{\partial \tilde{z}} + (\frac{\eta}{\rho v_{z,max} w}) \frac{\partial^2 \tilde{v_z}}{\partial \tilde{x}^2} + \frac{gw}{v_{z,max}^2} \tag{10}$$

We now have a dimensionless equation in (10). We can see a few dimensionless groups, including the new Pressure term and the spatial velocity term, and the scaled gravity term. We can see that when the constant $\frac{\eta}{\rho v_{z,max} w}$ is greater than 1, the $\frac{\partial^2 \tilde{v_z}}{\partial \tilde{x}^2}$ term dominates, and when its less than 1, the $\frac{\partial \tilde{P}}{\partial \tilde{z}}$ dominates. When both of these are the gravity term, it will dominate as a constant.

Problem 2

a. Given these ranges of values, is the flow of basaltic magma through Hawaiian dikes laminar, turbulent, or either one depending on certain parameters? Show how you arrived at your answer.

To check whether the flow is laminar or turbulent, we can check the limits of Reynolds number given by

$$Re = \frac{w \rho v_{z,max}}{\eta}.$$

We know that Re will be lowest when the Numerator is lowest, and the denominator is highest. Likewise we can check the max value when the numerator is largest, and the denominator is lowest. Checking these based on the given ranges of the parameters gives:

$$Re_{min} = \frac{(w \rho v_{z,max})_{min}}{\eta_{max}} = \frac{(0.5)(2400)(0.1)}{1000} = 0.12$$
$$Re_{max} = \frac{(w \rho v_{z,max})_{max}}{\eta_{min}} = \frac{(2)(2900)(1)}{10} = 580.$$

We can see that in both cases, the Re is below 2000 so the flow is laminar no matter the combination of allowed physical parameters.

b. Assuming everything else is the same aside from the dike width, do you think the flow of basaltic magma through these "giant" dikes was likely to be laminar, turbulent, or either one depending on certain parameters? Show how you arrived at your answer.

To predict if the giant dikes were turbulent or laminar, we can repeat the process in part A with the new width of 100m.

$$Re_{min} = \frac{(w \rho v_{z,max})_{min}}{\eta_{max}} = \frac{(100)(2400)(0.1)}{1000} = 24$$
$$Re_{max} = \frac{(w \rho v_{z,max})_{max}}{\eta_{min}} = \frac{(100)(2900)(1)}{10} = 29000.$$

Now we actually see a Re over 2000 so the flow can be either turbulent or laminar depending on the velocity, material density, and viscosity. However, if we look at the min value of the numerator we get $(w\rho v_{z,max})_{min} = (100)(2400)(0.1) = 24000$ which is greater than 2000. This means that the flow regime is very dependent on η.

Problem 3

a. Open the script flow_regime.m. Describe what each of the following lines of code do

See the code below for $v_{max} = 0.1$:)

```
% Comments with explanations line by line

clear all % clears all allocated variables, functions, breakpoints, ...
    etc from the current workspace
close all % closes all open figure windows
clc % clears the command window

% Vary magma viscosity
p = linspace(1,3,100); % exponent for magma viscosity
                       % Setting up a vector p with points between 3 and
                       % 1 with 100 grid points

eta = 10.^p; % magma viscosity (Pa s)
             % Setting eta to be 10^p coordinate

% Vary dike width
w = linspace(1,100,100); % Make another vector from x=1 to x=100
                         % with 100 grid points

% Constants
v = .1; % flow velocity (m/s) % Set V_max to be 0.1
rho = 2750; % magma density (kg/m^3) % Set density to be 2750
g = 9.8; % gravitational acceleration (m/s^2) % accelaration due to
    gravity

% Compute Reynold's number

[ETA,W] = meshgrid(eta,w); % Create a 2 2D matrices the size of eta x w
                           % ETA is a matrix where each row is the eta
                           % vector, with w rows, W has w on each row for
                           % eta columns

Re = rho*v*W./ETA; % Calculate Re as a 2D matrix using ETA and W

% Make plots
figure % creates a new figure window
```

```
contourf(log10(ETA),W,log10(Re)) % create contour plot of RE with Log(eta
    ) on the x
                                % W on the y.
xlabel('log viscosity (Pa s)') % Sets the x label
ylabel('Dike width (m)') % Sets the y label
set(gca, 'FontSize', 14) % set plot properties
colorbar % place a color bar
title(['log Re, v = ' num2str(v) ' m/s']) % set the title
caxis([0.5 4]) % set the color bar axis

% Extra parts for part b to add threshold line
hold on;
% Overlay a line for a specific Reynolds number threshold
threshold = 3.3; % Define a threshold for Reynolds number
contour(log10(ETA), W, log10(Re), [threshold threshold], 'k--', '
    LineWidth', 2); % Add contour line for threshold
```

Listing 1: flow_regime.m

b. Now run the script flow_regime.m. Print out a contour plot of Reynold's number as a function of the log magma viscosity and dike width. Draw the boundary for Re = 2000 (log(2000) = 3.3). Do this twice, one plot for velocity = 0.1 $\frac{m}{s}$ and another for velocity = 1$\frac{m}{s}$.

See plots below for $v = 0.1\frac{m}{s}$, and $v = 1\frac{m}{s}$.

Figure 1: Contour Plot of Dike width (m) vs log(Viscosity) (Pa s) with log(Re) shaded for $v_{max} = 0.1\frac{m}{s}$. Dashed line shows log(Re = 2000) = 3.3 .

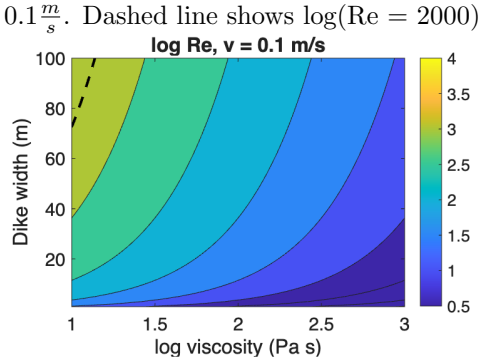
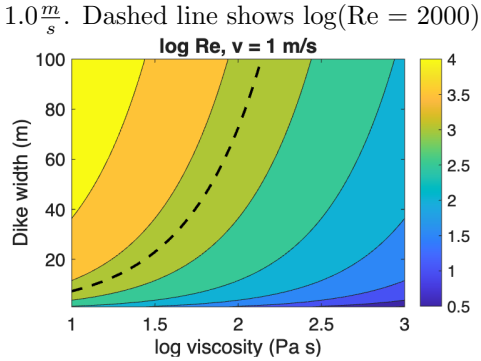


Figure 2: Contour Plot of Dike width (m) vs log(Viscosity) (Pa s) with log(Re) shaded for $v_{max} = 1.0\frac{m}{s}$. Dashed line shows log(Re = 2000) = 3.3 .



Problem 4

The Navier-Stokes equation is given as:

$$\rho \frac{\partial v_i}{\partial t} = -\frac{\partial P}{\partial x_i} + \eta \frac{\partial^2 v_i}{\partial x_k \partial x_k} + \rho g_i \tag{11}$$

a. Write out in component form (using x,y,z) all 3 equations represented by the indicial form of Navier-Stokes (eq 2). Remember that when an index appears twice in a term, that index is understood to take on all the values of its range, so the resulting terms are summed.

We can write out (11) in component form as:

$$\rho \frac{\partial v_x}{\partial t} = -\frac{\partial P}{\partial x} + \eta(\frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2}) + \rho g_x \tag{12}$$

$$\rho \frac{\partial v_y}{\partial t} = -\frac{\partial P}{\partial y} + \eta(\frac{\partial^2 v_y}{\partial x^2} + \frac{\partial^2 v_y}{\partial y^2} + \frac{\partial^2 v_y}{\partial z^2}) + \rho g_y \tag{13}$$

$$\rho \frac{\partial v_z}{\partial t} = -\frac{\partial P}{\partial z} + \eta(\frac{\partial^2 v_z}{\partial x^2} + \frac{\partial^2 v_z}{\partial y^2} + \frac{\partial^2 v_z}{\partial z^2}) + \rho g_z \tag{14}$$

b. Recall the definition of the del operator $\nabla = \hat{e}_i \frac{\partial}{\partial x_i}$ (also known as the gradient when applied to a scalar value). Write out ∇P in component form.

Since P is a scalar we can write out component for of ∇P as:

$$\nabla P_x = \frac{\partial P}{\partial x} \tag{15}$$

$$\nabla P_y = \frac{\partial P}{\partial y} \tag{16}$$

$$\nabla P_z = \frac{\partial P}{\partial z} \tag{17}$$

c. For the x-component of velocity v_x (or v_1), write out the Laplacian $\nabla^2 v_x = \nabla \cdot \nabla v_x$

We can write out $\nabla \cdot \nabla v_x$ as:

$$\nabla^2 v_x = \nabla \cdot \nabla v_x \tag{18}$$

$$= \nabla \cdot (\frac{\partial v_x}{\partial x}\hat{x} + \frac{\partial v_x}{\partial y}\hat{y} + \frac{\partial v_x}{\partial z}\hat{z}) \tag{19}$$

$$= \frac{\partial^2 v_x}{\partial x^2} + \frac{\partial^2 v_x}{\partial y^2} + \frac{\partial^2 v_x}{\partial z^2} \tag{20}$$

d. Using your results from b) and c), rewrite the Navier-Stokes equation above (eq 2) using vector notation.

Using our work above we can now rewrite (11) - (14) more compactly as

$$\rho \frac{\partial v_i}{\partial t} = -\frac{\partial P}{\partial x_i} + \eta \frac{\partial^2 v_i}{\partial x_k \partial x_k} + \rho g_i \tag{21}$$

$$\rho \frac{\partial \bar{v}}{\partial t} = -\nabla P + \eta \nabla^2 \bar{v} + \rho \bar{g}. \tag{22}$$

$$\tag{23}$$